Harsh Pratap Singh

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EDUCATION

DIT UNIVERSITY (Dehradun) — **Aug 2023 - May 2027**

Bachelor of Technology in Mechanical Engineering — CGPA: 8.61/10

Sr. Secondary — 2023/PCM– 76.6%

Secondary — 2021 - 84 %

EXPERIENCE / INTERNSHIPS

1.Research Assistant (DIT UNIVERSITY) (Link) — Feb 2025 - Jun 2025

- Assisted in experimental research on **heat exchanger performance** — *On-site*
- Conducted data collection and thermal analysis (Nusselt number, friction factor)
- Evaluated **heat transfer enhancement using triangular winglets**

2.Uttarakhand Transport Corporation (Dehradun) (Link) — Jun 2025

- Hands-on experience in **vehicle inspection and maintenance** — *On-site*
- Studied **IC engines, braking, and cooling systems**

3. Indian Institute of Remote Sensing (Dehradun) (Link) — Aug 2025 - Nov 2025

- Training in **Remote Sensing, GIS, and GNSS** — *Remote*
- Learned **geospatial data analysis and mapping**

4. Project Intern Institute for Plasma Research (IPR) (Gandhinagar) (Link) — May 2026 - Jul 2026

- Interned in the **Spherical Tokamak Division** — *On-site*
- Performed **FEA-based thickness optimization of a rectangular SS304L vacuum vessel flange**
- Conducted **transient thermal (welding simulation)** and **static structural analysis** in **ANSYS Mechanical**

ACHIEVEMENTS / CERTIFICATIONS (Certificates)

- Completed Summer School Program (SSP) 2026 (IPR Gandhinagar)
- Research paper presented at **INCOTHERM 2025** [IIT (ISM) Dhanbad]
- Certified in **ANSYS Fluent** (Ansys)
- Completed **ANSYS Technical Training** (CRAFT)
- Completed **MATLAB & Simulink Onramp** (MathWorks)
- Certified in **AutoCAD and Fusion 360** for design and modeling
- Completed **Industrial Automation & Robotics Training** (DIT University)

DOMAIN / TECHNICAL SKILLS

1. **Technical Skills:** AutoCAD (2D & 3D), Fusion 360, ANSYS Fluent (CFD), ANSYS Mechanical (FEA), MATLAB, Simulink, Python, C, Excel, Hydraulics & Pneumatics.

2. **Hands-on & Workshop:** Vehicle inspection, maintenance & fault diagnosis, Thermal system analysis & experimentation, Industrial automation & hydraulic systems.

3. **Soft Skills:** Team collaboration, Technical documentation, Problem-solving

PROJECTS

1.Thermal performance of a heat exchanger tube equipped with strip insert with staggered triangular winglets (Published Patent IN202611037015 A1 & Conference Proceedings) (Link)

- Investigated strip inserts with staggered triangular winglets
- Focused on heat transfer enhancement and pressure drop analysis

2.Heat transfer enhancement in a cylindrical tube equipped with inline triangular winglets in strip insert (Conference Proceedings)

- Conducted experimental analysis on heat transfer and fluid flow behavior
- Evaluated thermal performance parameters under different configurations

3. Modelling & CFD simulation of lithium-ion battery pack (Link)

- Developed and analyzed a 16-cell lithium-ion battery pack using ANSYS Fluent
- Studied temperature distribution and cooling effectiveness

4. Modelling and FEA of wall wheel cart caddy (Link)

- Performed structural analysis and stress optimization

5. FEA-Based Design Optimization of Rectangular Vacuum Vessel Flange for Spherical Tokamak (Link)

- Compared 4 plate thicknesses (13/16/19/22 mm) × 3 welding sequences to minimize weld-induced distortion
- Selected 16 mm flange with symmetric welding sequence as optimum, cutting max von-Mises stress ~13% vs 13 mm baseline (SF ≈ 2.0)

LEADERSHIP/EXTRACURRICULAR

1.Secretary, SAC Club (Students' Automotive Club), DIT University:

- Managed event coordination
- Documentation
- Inter - department communication;
- Promoted from member to secretary based on performance

2.Industrial Visit, Hero MotoCorp, Haridwar:

- Gained insights into large-scale automobile manufacturing
- Assembly line processes
- Quality control systems

3. Industrial Visit, ONGC, Dehradun:

- Observed real-world mechanical and petroleum systems
- Learned about thermal and energy management operations

4.Industrial Visit, ISRO Space Applications Centre (SAC), Ahmedabad:

- Gained exposure to Remote Sensing, GIS and satellite applications.
- Attended sessions on satellite image processing and enhancement techniques.
- Observed MATLAB applications in image analysis and geospatial data processing.
- Learned about semiconductor technologies and space electronic systems.
- Developed understanding of communication, navigation and Earth observation satellites